

Vishal R

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Summary

Aspiring Software Engineer and Full Stack Developer Intern (Computer Science student) with hands-on experience building scalable, production-style web and AI-integrated applications using React.js, Next.js, Node.js, Express.js, and PostgreSQL. Experienced in REST API design, database schema design, authentication, authorization, and LLM API integration (OpenAI, Gemini). Currently architecting a 4-metric IoT-based defence health-monitoring system. Strong foundation in system design and version control.

Technical Skills

Languages: JavaScript, TypeScript, Python, Java, SQL, HTML5, CSS3

Frontend: React.js, Next.js 14, Tailwind CSS, shadcn/ui, Framer Motion, Responsive Design

Backend: Node.js, Express.js, REST API Design, JWT Authentication, Role-Based Access Control (RBAC), Flask

Database: PostgreSQL, Supabase, MySQL, Database Schema Design, Row-Level Security

AI, APIs: OpenAI API, Gemini API, Prompt Engineering, LLM Integration

Tools, Practices: Git, GitHub, Docker, Vercel, Netlify, VS Code

Projects

RepoLens AI — AI Repository Analysis Platform (Personal Project)

Completed

Next.js, React, TypeScript, Tailwind CSS, OpenRouter API, Vercel

- Built an AI-powered platform that analyzes GitHub repositories and explains unfamiliar codebases in a simple, developer-friendly manner.
- Designed an automated analysis engine to extract project architecture, technology stack, workflows, and repository structure.
- Implemented secure server-side AI processing with protected API architecture and detailed repository insights.
- Developed repository insights including project purpose, learning path recommendations, difficulty assessment, and improvement suggestions.
- Developed a conversational AI assistant for repository-specific Q&A and detailed codebase insights.

AI Trip Planner

Completed

Flutter, Supabase, OpenAI API, Gemini API, PostgreSQL

- Built a cross-platform mobile application that generates personalized travel itineraries based on user preferences, budget, and duration.
- Designed a modular 3-tier architecture using Flutter, Supabase, and AI services for scalable integration.
- Engineered structured AI response handling to ensure reliable itinerary generation and seamless UI rendering.
- Implemented secure authentication, row-level security, and real-time itinerary management features using Supabase.

Smart Military Vest — IoT Defence System

In Progress

ESP32, Sensors, LoRa, GSM, GPS

- Developing a defence-grade wearable system for real-time monitoring of soldier health metrics including heart rate, SpO2, and temperature.
- Architecting dual-channel communication using LoRa and GSM with encrypted data transmission for reliability in low-connectivity environments.
- Building an automated distress alert system with sensor-driven monitoring and emergency notification capabilities.

Education

Bachelor of Engineering in Computer Science and Engineering

Expected 2027

K.S. Institute of Technology, Bangalore

Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming, System Design

Languages

English, Kannada, Hindi, Telugu (Native), Tamil (Fluent)